

Gerald Mahapranaja Pillian

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Professional Summary

Results-driven Information Systems student with a strong foundation across cybersecurity governance, full-stack software development, and data analytics. Demonstrated leadership through international academic exchange at NTUST Taiwan, published research spanning computer vision and educational data mining, hands-on enterprise application development for IT governance at PT. PAL Indonesia, and multi-disciplinary project involvement across engineering, marketing, and event management.

Education

National Taiwan University of Science and Technology (NTUST)

August 2025 – January 2026

Exchange Program – Department of Computer Science

Taipei, Taiwan

- Developed project: **Fuzzy Logic Control for Dynamic Target Tracking in Two-Axis Autonomous Aiming Systems**
- Developed project: **Email Communication Simulation System Designs**
- Developed ML project (Machine Learning Course): **Particle Swarm Optimized Neural Networks for Real-Time Air Quality Forecasting in Taipei's Urban Microenvironments** – trained on **100,000+** samples

Institut Teknologi Sepuluh Nopember (ITS)

August 2022 – October 2026

Bachelor of Information Systems

Surabaya, Indonesia

- Final Cumulative GPA: **3.56/4.0**, Predicate: **Sangat Memuaskan** (Very Satisfactory) | Runner-up FTEIC 2022 BPC | Semi-Finalist UGMBCC (2023) | Semi-Finalist DANAMON DIBPC (2024)

Lazuardi High School

October 2019 – October 2022

Natural Sciences (IPA) Track

Indonesia

- Graduated with average **89.03**; ranked **7th** in cohort; represented school in international academic competitions

Experience & Involvement

Information Technology Intern – Cybersecurity & IT Governance

August 2026 – Present

PT. PAL Surabaya, Information Technology Division

Surabaya, Indonesia

- Worked within the Cybersecurity and Governance/IT Governance department, primarily on the Governance team, drafting **SOPs** and **COBIT 2019** domain assessments to identify areas for improvement across the company's governance domains
- Independently architecting and developing **5 full-stack internal platforms in Laravel**, applying secure coding practices (RBAC, input validation, audit logging) and production-grade code structure:
 - * **Cybersecurity Finding & Remediation Tracker** – logs audit/pentest findings and tracks remediation against SLAs
 - * **Security Exception & Risk Register** – manages risk-acceptance requests and exception approval workflows
 - * **Internal IT Service Dependency Map** – visualizes system/service interdependencies for change-impact analysis
 - * **"Lessons Learned" Engineering Knowledge Base** – centralizes post-incident and post-project documentation for reuse
 - * **Internal "Process Friction" Reporter** – captures and triages operational friction points reported by staff

UI/UX Designer & Developer

May 2024 – June 2024

Data Engineering and Business Intelligence Laboratory, IS ITS

Surabaya, Indonesia

- Designed end-to-end UI/UX for **CLASTIC**, an AI-powered plastic waste sorting mobile app; conducted field testing in Sumberbening Village
- Implemented web and graphic design across digital platforms, ensuring cohesive and user-friendly experiences

Expert Staff – External Department

April 2023 – August 2025

Information Systems Expo (ISE!), ITS

Surabaya, Indonesia

- Progressed from General Staff to Expert Staff; managed and mentored a team of 3 staff members
- Pitched industry professionals as guests; served as **Master of Ceremonies** for UX Academy and Data Software Academy
- Achieved a KPI of **300+ participants** across organized programs

Student Staff Volunteer – ITS Global Engagement

December 2023 – February 2024

Institut Teknologi Sepuluh Nopember

Surabaya, Indonesia

- Liaised for **100+ international students** across **10+ major events**; served as speaker, MC, logistics coordinator, and guide

Assistant Lecturer – Introduction to Information Systems Concept

September 2023 – December 2023

Department of Information Systems, ITS

Surabaya, Indonesia

- Designed assignments and projects; provided grading, academic support, and coordination with course lecturers

Lab Assistant, Teaching Assistant & Grader – Pemodelan Analitika Prediktif

March 2025 – July 2025

Department of Information Systems, ITS

Surabaya, Indonesia

- Supported lab sessions and course instruction; graded assignments and provided academic support to students

Teaching Assistant & Grader – Introduction to Statistics

October 2023 – December 2023

Department of Information Systems, ITS

Surabaya, Indonesia

- Assisted course instruction and graded student assignments and assessments

Strategy & Development Staff

June 2023 – February 2024

IEEE, ITS Surabaya Student Branch

Surabaya, Indonesia

- Principal editor for marketing and archival content; completed **5+ major projects** on the Documentation and Editorial Team

Research & Publications

- Published Paper | *ResearchGate*

2024

- “Understanding User Perceptions of Camera and Microphone Access in Mobile Applications: Exploring Permission Granting and Associated Security and Privacy Risks”
 - Investigated user trust patterns and security/privacy implications of app permission systems on mobile platforms
- Research Paper

2026

- “Implementation of Drone for Pipeline Visual Based Inspection Using YOLOv12n with GPS Based Localization”
 - Developed a drone-assisted visual inspection framework integrating **YOLOv12n** and **TensorFlow** for real-time detection of corrosion and third-party damage on pipelines, with **GPS-based geotagging** for spatial defect mapping and predictive maintenance
- Undergraduate Thesis

2026

- “Hierarchical Cluster Analysis of Medical Student Performance Across Multiple Assessment Modalities”
 - Applied Ward’s hierarchical clustering (Euclidean distance, Z-score standardisation) to 98 medical students across four assessment modalities at FKK ITS, identifying **3 statistically distinct performance clusters** (ANOVA, p < 0.001) to support early identification of at-risk students

Notable Projects

- Particle Swarm Optimized Neural Networks for Air Quality Forecasting

NTUST, 2025

- Built and trained a neural network on **100,000+ samples** using PSO for hyperparameter tuning to predict real-time urban air quality
- Fuzzy Logic Control for Dynamic Target Tracking in Two-Axis Autonomous Aiming Systems

NTUST, 2025

- Designed a fuzzy logic controller for real-time dual-axis target tracking; applied control theory and computational logic
- Email Communication Simulation System Designs

NTUST, 2025

- Engineered a simulation of email communication protocols; focused on system architecture and server-client flow modelling
- CLASTIC – AI-Based Plastic Waste Sorting Mobile Application

ITS, 2024

- Led UI/UX design for an AI-powered waste management app deployed in rural Indonesia; integrated ML-based plastic classification

Certifications

- SAP Certified Application Associate

Issued July 2024

SAP BusinessObjects BI Platform 4.2 | SAP University Alliances

Offline

 - 3-month ERP integration program covering **SD, MM, and PP** modules; signed by Mr. Crispian Tan, SAP University Alliances
- Data Analyst Certification

August – October 2023

MySkill

Online

 - Hands-on training in SQL, Python, and DBMS; built and presented a guided capstone data analysis project
- Cisco DevNet Associate

Issued July 2025

Cisco Networking Academy | Institut Teknologi Sepuluh Nopember

Offline

 - Certified in software-defined networking fundamentals, network automation, and REST API-based infrastructure programmability

Technical Skills

- Programming:

Python, SQL, JavaScript, HTML/PHP (Laravel), Java, VBA
- Data & ML:

R Studio, DBMS/SQL, Machine Learning (Neural Networks, PSO, YOLO/Computer Vision), SAP ERP & BI Platform
- Web & App Dev:

PHP/Laravel (secure, production-grade development), Supabase (Database, Auth, APIs), Google Cloud APIs, REST API Integration
- Networking & Security:

SDN Fundamentals (Cisco DevNet), COBIT 2019, Security Governance, Risk & Exception Management
- Tools:

Microsoft Office (Advanced), Google Workspace (Advanced), GDevelop, MicroStudio
- Design & Media:

UI/UX Design, Video Editing (Advanced), Professional Photography (Advanced), Photo Editing, General Marketing
- Soft Skills:

Leadership, Public Speaking, Event Management, Problem Solving, Adaptability, Emotional Intelligence

Languages

- Indonesian – Native

Javanese – Intermediate

Arabic – Basic

German – Basic
- English – Fluent

Mandarin – Basic

French – Basic